Case Study 5: Repeated Transaction Detection

Assignment Case No. 28 | Level: Beginner

01 PROBLEM STATEMENT

Process transaction identifiers and amounts in a list, detect duplicate transactions, identify threshold violations, and generate a list of suspicious entries.

Problem interpretation: A transaction log may contain duplicate identifiers or unusually large amounts. This program finds duplicate IDs and flags amounts above a specified threshold.

02 OBJECTIVES

Process transaction records stored as tuples.

Detect duplicate transaction IDs.

Flag transactions above a threshold.

Build and display a suspicious-entry list.

03 ASSUMPTIONS

Each transaction is a tuple of (ID, amount).

IDs should be unique.

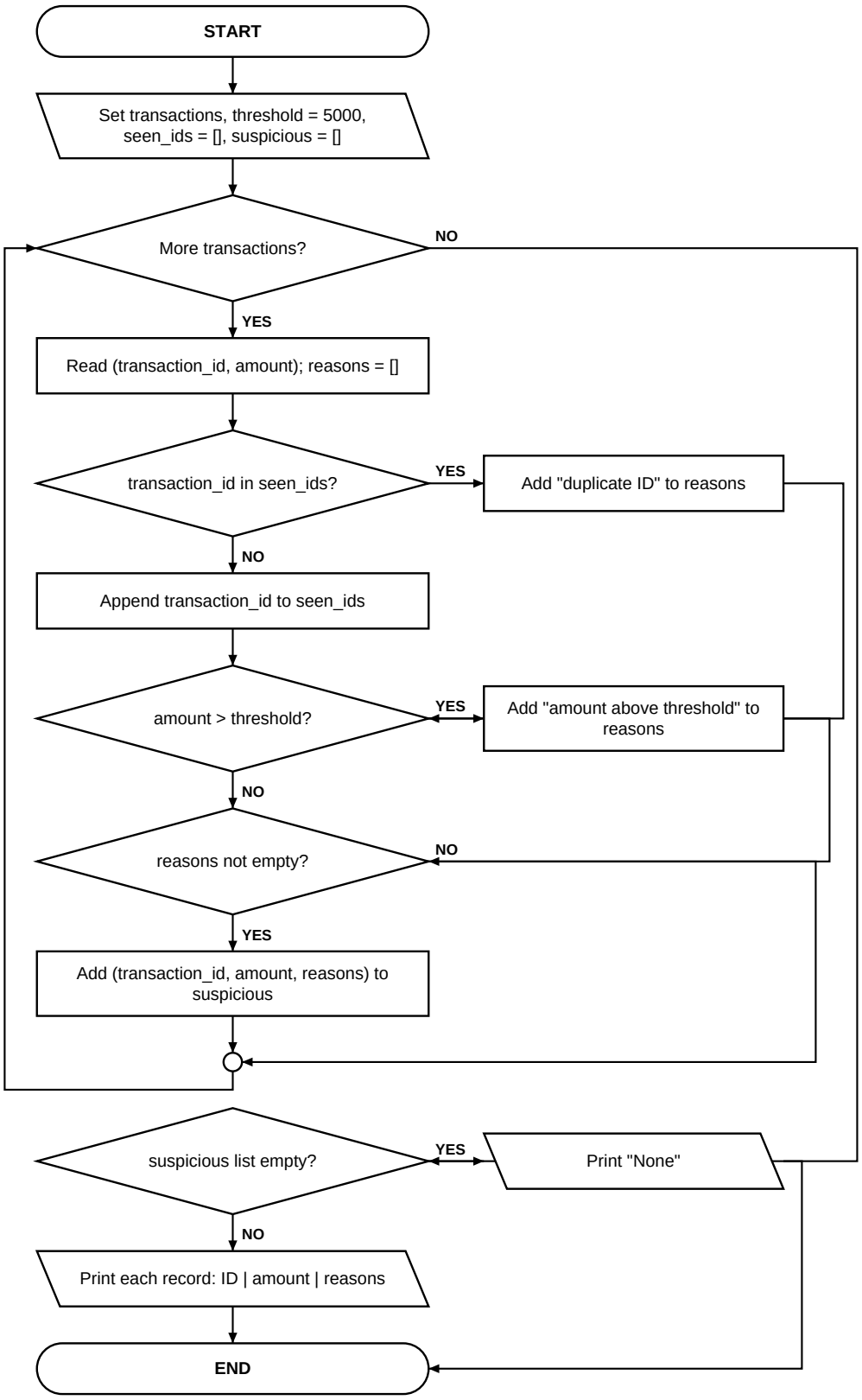
An amount above ₹5,000 is flagged for review.

A flagged transaction is suspicious for review, not proof of fraud.

04 ALGORITHM

1. Start.
2. Store transactions as tuples in a list.
3. Create an empty list for seen IDs and suspicious records.
4. For each transaction, check whether its ID has appeared before.
5. Also check whether its amount exceeds the threshold.
6. Add any record meeting either condition to the suspicious list.
7. Display the suspicious records.
8. Stop.

05 FLOWCHART (TEXT FLOW)



06 PSEUDOCODE

```
BEGIN
    transactions = [(T101, 1200), (T102, 6500), (T101, 1200), (T104, 800), (T105, 7200)]
    threshold = 5000; seen_ids = []; suspicious = []
    FOR each (transaction_id, amount) IN transactions
        reasons = []
        IF transaction_id IN seen_ids THEN
            add "duplicate ID" to reasons
        ELSE
            add transaction_id to seen_ids
        END IF
        IF amount > threshold THEN add "amount above threshold" to reasons
        IF reasons is not empty THEN add (transaction_id, amount, reasons) to suspicious
    END FOR
    PRINT "SUSPICIOUS TRANSACTIONS FOR REVIEW"
    IF suspicious is not empty THEN
        FOR each record in suspicious: PRINT ID | amount | reasons
    ELSE
        PRINT "None"
    END IF
END
```

07 PYTHON SOURCE CODE

The complete Python program is provided in the separate file 05_Repeated_Transaction_Detection_Code.pdf.

08 TEST CASES

INPUT / SCENARIO	CHECK	EXPECTED RESULT
T101 appears twice	Second T101 repeats an ID	Flagged: duplicate ID
T102, ₹6500	Above ₹5000	Flagged: amount above threshold
T105, ₹7200	Above ₹5000	Flagged: amount above threshold
T104, ₹800	Unique and below threshold	Not flagged
Amount exactly ₹5000	Boundary: not above threshold	Not flagged

09 SAMPLE OUTPUT PREVIEW

```
SUSPICIOUS TRANSACTIONS FOR REVIEW
T102 | ₹6500 | amount above threshold
T101 | ₹1200 | duplicate ID
T105 | ₹7200 | amount above threshold
```

10 CONCLUSION

The program applies list membership checks and tuple unpacking to find duplicate identifiers and high-value transactions for review.